

Xin HUANG

Hong Kong

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Education

University of Hong Kong,

Hong Kong *Sept 2025 - Jul 2026*

M.Sc. (Computer Science)

School of Computing and Data Science, HKU

Course:

Natural Language Processing, Deep Learning, Data Mining, Introduction to Unmanned System, Digital Signal Processing and Computer Vision

University of Hong Kong,

Hong Kong *Sept 2021 - Jul 2025*

B.Sc. (Major: Mathematics; Minor: Computer Science)

Dean's Honours List in faculty of Science, HKU

Cumulative GPA: 3.64/4.3 First class honor graduate (about 10% of BSc)

Course:

Mathematics: Operations Research, Probability and Statistics, Scientific Computing, Differential Equations, Optimization, Numerical analysis, Financial Calculus, Game Theory, Functional Analysis, Linear algebra I & II;

Computer Science: Algorithm Design, Data structure and algorithm, Artificial Intelligence, Database Management

Experience

AI Model & System Research Intern,

Hong Kong *Sept 2025 – Present*

Centre for Artificial Intelligence and Robotics (CAIR),

Hong Kong Institute of Science & Innovation, Chinese Academy of Sciences

- Participated in research on continual learning algorithms for medical scenarios, assisting in addressing long-tail and forgetting problems.
- Assisted in continual fine-tuning and reinforcement fine-tuning methods for MLLMs, including model architecture improvements and training strategy experiments.
- Conducted experimental analysis and data organization & Supported the drafting of technical documents and research papers.

Algorithm Intern, ZETA VISTA,

Shenzhen, China *Jul 2025 – Aug 2025*

IT Department

- Designed and developed a child face recognition and classification application leveraging FaceNet, enhancing photo-classification efficiency on the Family-School interaction platform.
- Addressed high facial similarity challenges by implementing a pre-recognition image quality evaluator, utilizing an Inception-ResNet V1 model pre-trained on VGGFace2 and CASIA-Webface as the feature extractor, and applying adaptive thresholding.
- Engineered an AI Agent Workflow based on DIFY to automate photo album generation.

Research Assistant, University of Hong Kong,

Hong Kong *Nov 2024 – Mar 2025*

HKU Business School

- Analyzed complex mathematical problems and formalized proofs for theorems/lemmas within operations management models, optimizing contracts and strategic decisions under constraints.
- Employed Linear Programming and optimization techniques (e.g., KKT conditions, Gradient Descent) to solve resource allocation and supply chain challenges.
- Enhanced expertise in mathematical modeling and data-driven decision-making.

Algorithm Research Intern, University of Alberta,
Experiments with Graph Signal Processing

Alberta, Canada Jun 2024 – Aug 2024

- Investigated the feasibility of replacing Multi-Layer Perceptrons (MLPs) with Kolmogorov-Arnold Networks (KANs) – utilizing learnable univariate nonlinear functions instead of linear weights – through data analysis and experimental evaluation.
- Conducted systematic Fourier and Wavelet analysis on user-product interaction data for recommendation system; compared B-spline, Fourier, and Wavelet bases within the KAN model, demonstrating wavelet bases' superior performance in enhancing model expressiveness.
- Explored integrating hypergraphs with the KAN architecture to model complex user-product relationships, aiming to develop a recommendation system outperforming classical models (e.g., NGCF, LightGCN).

Research Project

Efficient numerical schemes for Path-dependent PDEs

Feb 2025 - May 2025

- Studied the efficient numerical algorithms for Path-dependent Partial Differential Equations in non-Markov process modeling to price financial derivatives.
- Conducted readings and duplication on the existing extended theorems on Path-dependent PDE with monotone schemes for fully nonlinear PPDE, including finite difference method and probabilistic scheme.

Achievement

- 2021-2022: HKU Entrance Full Scholarship
- 2022-2023: Dean's Honours List; HKU Full Scholarship for Outstanding Academic Talents
- 2023-2024: Dean's Honours List; HKU Full Scholarship for Outstanding Academic Talents
- 2024-2025: HKU Full Scholarship for Outstanding Academic Talents; First-class Honour Graduate

Skill

- Python (PyTorch, Numpy, Pandas), C++, Vue.js, SQL, Tableau
- English (IELTS 8), Mandarin, Cantonese